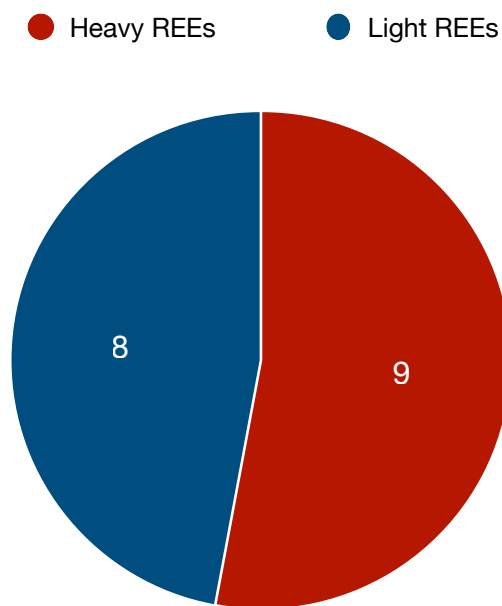


Assignment - 1

Rare Earth Elements

Rare Earth Elements, also known as REEs, are group of seventeen chemically similar elements in Periodic Table. The reason why they are so interesting is also the reason they are harder to work with - which is that they are almost chemically identical to each other. REEs consists of elements with atomic number 57 all the way upto 71 + Scandium (atomic number 21) and Yttrium (atomic number 39). Both of these elements behave a lot like lanthanides, with Yttrium founding in the same geological deposits as other rare earth elements. These REEs are divided into two categories: Light Rare Earth Earth Elements and Heavy Rare Earth Elements



Classification of 17 Rare Earth Elements

Light Rare Earth Elements (LREEs)

The light rare earths comprises of elements ranging from atomic number 57 to 64. These elements are available plentiful in Earth's crust and are generally more easier to extract from the ground, often found in deposits like bastanites and carbonatites.

Because these, are relatively more abundant, they tend to be cheaper and more readily available.

Heavy Rare Earth Elements (HREEs)

Heavy rare earths have heavier elements running from atomic number 65 to 71. These elements are much scarcer in nature and harder to extract, making them valuable as they are also very useful. Geologically, they often found in ionic adsorption clays rather than like LREEs. The difficulty in finding and processing them is a very big reason why the heavy rare earth elements command such prices in the global market.

Scandium and Yttrium are usually discussed separately because they are not lanthanides, but they are included in REE group.

Importance: Why Rare Earth Matter so Much?

The reason rare earth are so important is due to their electronic configuration. They have unfilled 4f electron subshells, which gives them several very unique properties. Those properties include luminescence, they can exhibit magnetic behaviour, and other useful optical properties that allows them to be used in Optics industry. These properties make them essential for huge range of modern technologies, ranging from smartphone screens to wind turbines.

One of the most important uses of rare earth elements is in making Permanent Magnets. Elements like Neodymium and Samarium are used to make strong permanent magnets that can generate high intensity magnetic fields while being relatively small and lightweight. These magnets play key role in Electric Vehicle motors, wind turbine generators, speakers, hard disk drives etc.

Rare earths also play a critical role in glass polishing, petroleum cracking catalysts, and even in medical imaging. Gadolinium is used as contrast agent in MRI scans in hospitals. Lanthanum is used in camera lenses and high-refractive-index glass. Cerium is used in automobile catalytic converters to reduce emissions. The reason why they are so significant that there is no substitute for these elements in most of their applications, which is also makes them important and commercially valuable.

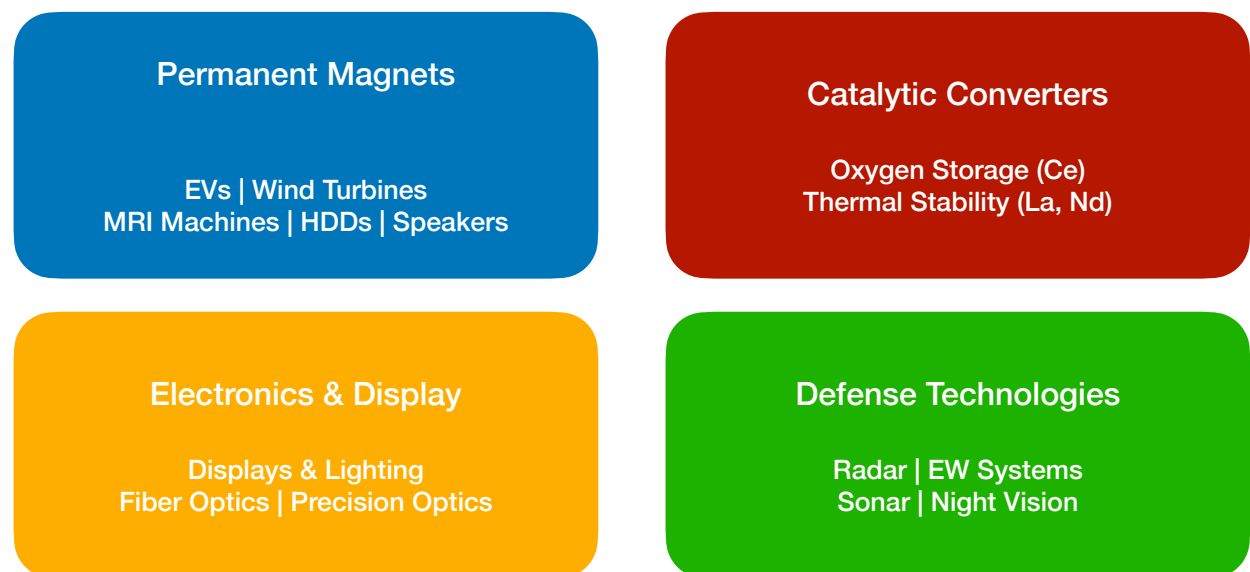
And due to these diverse use-cases of REEs, the current supply situation is concerning. Demand is keep rising through multiple industries at once, while the supply is limited to very few countries. Any disruption, whether political, economical or environmental, can cause a bottleneck in manufacturing.

Uses: Permanent magnets (Neodymium-Iron-Boron and Samarium-Cobalt)

- 1. Neodymium-Iron-Boron** - It has complex tetragonal structure and is intermetallic compound. There are two forms: Sintered $\text{Nd}_2\text{Fe}_{14}\text{B}$, which is produced through high-temperature powder compaction to achieve maximum magnetic strength. And bonded $\text{Nd}_2\text{Fe}_{14}\text{B}$, which is made by blending magnet powder with polymer binders. They deliver exceptional magnetic strength with high energy density, however they have downsides which is that they require protective coating to prevent oxidation and they need rare earth additives to operate effectively at temperatures above 150 degree Celcius. This forms a real temperature constraint in many high-performance applications.
- 2. Samarium-Cobalt** - Samarium-Cobalt magnets comes in two primary chemical forms: SmCo_5 , which is 1:5 composition of Samarium and Cobalt, and

Sm₂Co₁₇, which has 2:17 composition. They are slightly less powerful than neodymium magnets, but they have very good advantages. Some of these advantages are that they resist corrosion exceptionally well, and maintain its magnetic strength at extreme temperatures - ranging from 300 to 550 degrees Celsius - without needing to add heavy rare earth additives. This advantage makes them suitable for aerospace and military applications where responsiveness and reliability at extreme conditions such as high temperature, vibrations, and radiations is critical.

These magnets are used in various industries such as Electric vehicles, Renewable Energy Infrastructures, Industrial Automation, Robotics, Heating Ventilation and Air Conditioning (HVAC) systems etc.

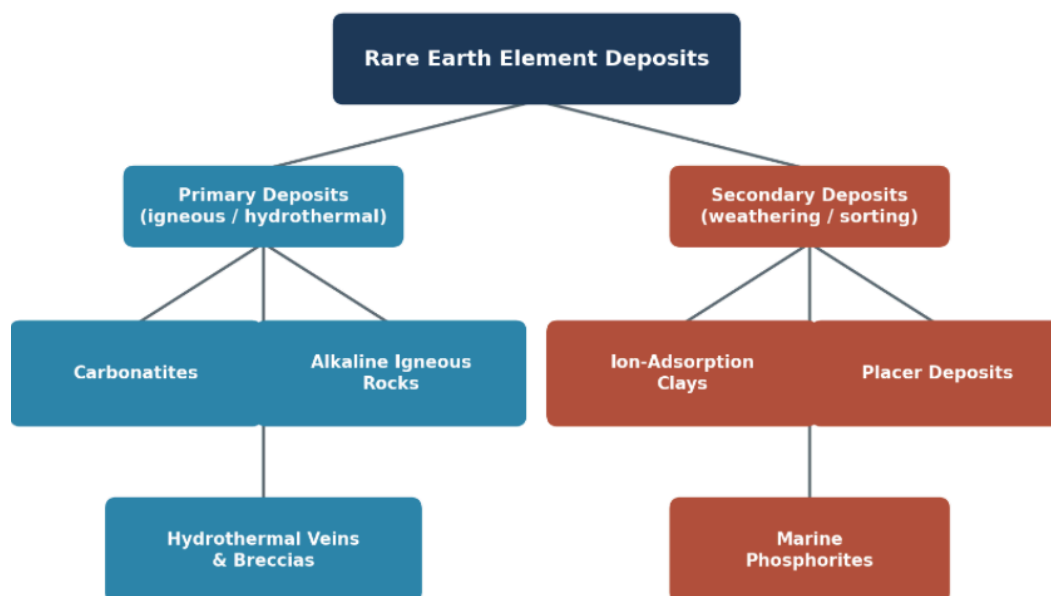


Major Applications of Rare Earth Elements

Occurrences: Where do Rare Earths Come From?

Rare Earth elements are extracted from deposits which can be classified into two main categories: Primary Deposits and Secondary Deposits.

- 1. Primary Deposits** - Primary deposits are hard rock formations created by igneous and hydrothermal processes. | Basically minerals are formed due to crystallisation directly from cooling magma or hot fluids, flowing through the earth's crust. There are few types of primary deposits. Carbonatites are one type. These are rocks that are made up of at least 50% primary carbonate minerals. They are world's main source of light rare earth elements. Also Alkaline Igneous Rocks are also another source of REEs, which form from silica-poor magmas. These rocks are rich in alkali metals like Sodium and Potassium, and they tend to be rich in heavy rare earth elements and Yttrium, along with many useful metals like Zirconium and Niobium.



Classification of REE deposits into Primary and Secondary

- 2. Secondary Deposits** - These rocks form when primary igneous rocks get broken down by natural processes like Erosion, weathering, and transport. It is due to

mechanical sorting and chemical weathering that concentrate the rare earths into extractable resources, instead of magmatic heat. For example, Ion-Adsorption clays are formed in subtropical climates where granite rocks undergo intense chemical weathering. The rare earth leach out of the primary minerals and absorbed onto the surfaces of clay minerals as loosely bounded ions. Placer deposits are another type. These form through physical weathering and mechanical sorting of heavy minerals in riverbeds, deltas, and coastal beach sands.

Production: How Are They Extracted?

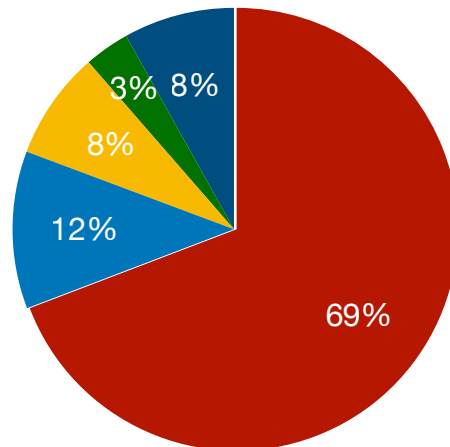
Producing rare earth elements is a very complex process because they all occur in same mineral deposits and have similar chemical properties due to which separating them is very difficult. The process starts with mining ores from carbonatites, alkaline rocks, or ion-adsorption clays. Once the ores are extracted, they undergo through a process to increase concentration of rare earth-bearing mineral, called Beneficiation methods.

After beneficiation, the concentrated ores are treated with acid or alkaline leaching. This chemical treatment process dissolves the rare earths, after which they have to be separated from one another. Solvent extraction and Ion exchange methods are used for this separation process. These are multi-step processes that require a lot of precision and also generates a fair amount of waste. After the separation process, these rare earths are converted to compounds like rare earth oxides etc, and then these

are processed into metals and alloys. After this processing, these products can be finally made into permanent magnets, catalysts, and other advanced materials.

China is currently the most dominating producer and processor of rare earth elements. It has a large share of the world's mining, separation, refining and producing permanent magnets. United States, Australia, Myanmar, and several other countries also contribute to this chain, but many of them are still dependent on China for further processing. This domination in the supply chain creates a major concern because it is highly concentrated in a single country, China, and any disruption in it can affect the industries all around the world, as REEs' uses are highly diversified in multiple industries as mentioned above.

● China ● United States ● Myanmar
● Australia ● Res of the World



**Global Rare Earth Elements
Production by countries**

Current Demand for Permanent Magnets

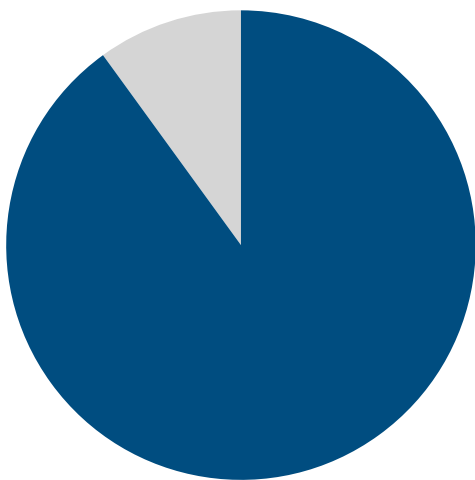
While rare earths have many applications, permanent magnets are most significant. They account for a major share of the total REE market. The Neodymium-Iron-Boron types represent the fastest growing segment of rare earth market. Due to the shift

towards electrification, automation and digital technologies has created a high demand of these magnets. According to IEA, demand for key magnet rare earths has doubled since 2015 and crossed 90 kT in 2024. IEA predicts the demand to grow by 30% or more by 2030 under today's policy settings. Industries driving these demands are:

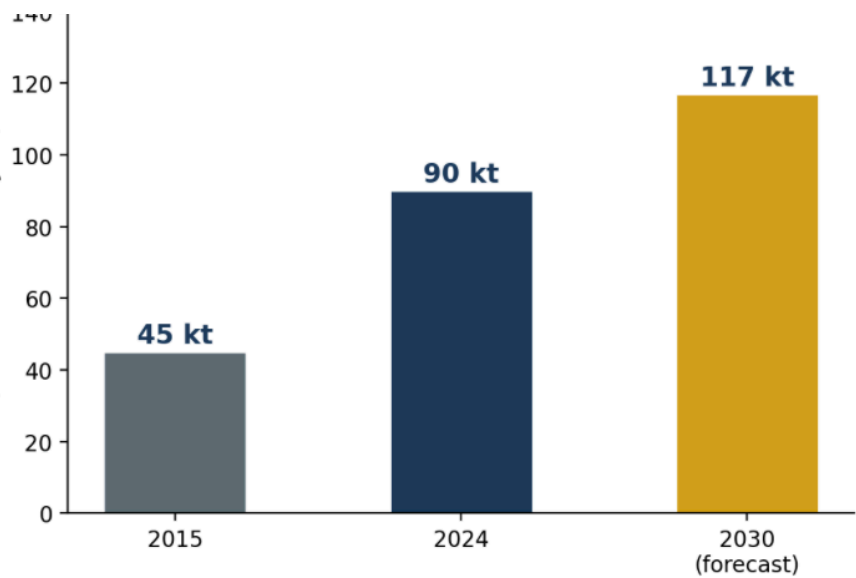
- 1. Electric Vehicles** - Modern EV heavily rely on permanent magnets synchronous motors, for high efficiency, compact size and exceptional power-to-weight ratios. A typical EV uses about 1 to 2 Kg of NdFeB magnet material for its primary drive motor.
- 2. Renewable Energy** - Offshore and advanced onshore wind system have shifted towards Direct-Drive Permanent Magnet Generators. This eliminates traditional heavy gearboxes and reduce maintenance. A single generator requires around 200 to 650 Kg of NdFeB magnets per megawatt of output.
- 3. Industrial Automation and Digital Infrastructure** - Precision robots, CNC equipment and servo motors depend on permanent magnets to ensure fast response and precise positioning. In Heating, Ventilation and Air Conditioning (HVAC) systems, NdFeB magnets helps to achieve strict energy-efficiency standards. And in data centres, cooling pumps, server fan motors, and backup power equipment also consume increasing quantity of these magnets.

Share of REE Market: Permanent Magnets vs other applications

Permanent magnets All other applications



Global Demand for Magnet Rare Earth Elements IKE



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